UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. Interlacing displays every line of a video frame at the same time.
2. Sampling converts a continuous sound wave into digital values.
3. A storyboard is a visual plan of a multimedia sequence.
4. RAM is a type of permanent storage.
5. A linear presentation allows users to choose their own path.
6. MIDI files store digitized audio samples.
7. Hertz measures sound frequency.
8. Hypermedia combines multimedia with hyperlinks.
9. Card-based authoring tools are best for messages with a clear beginning and end.
10. Cel animation is created by drawing each frame by hand.
11. A prototype is an early model of a product.
12. Tweening creates intermediate frames between keyframes.
13. RGB is an additive colour model.
14. A web browser displays hypertext documents.
15. Path animation moves an object along a predefined route.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Storyboard	B. Sampling	C. Card-based tool
D. MIDI	E. Tweening	F. Vector
G. Plotter	H. Multimedia	I. Chroma key
J. Scanner	K. Keyframe	L. Linear multimedia
M. Frame rate	N. Serif	O. Bitmap

1. important frame in an animation
2. integration of text, graphics, audio, video and animation
3. image made of pixels
4. generating in-between frames
5. input device that digitises documents
6. visual sequence plan
7. output device for large drawings
8. converting sound into digital values
9. image made of mathematical objects
10. presentation without user control of sequence
11. frames displayed per second
12. standard for musical notes and instructions
13. authoring tool organised like pages or cards
14. font with small strokes at character ends
15. replacing a colour background

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Tweening creates:
   A. In-between frames
   B. Text fonts
   C. Video codecs
   D. Sound effects

2. Card-based authoring tools are suited to:
   A. Messages with a beginning and end
   B. Live video only
   C. Audio editing only
   D. 3D modelling only

3. Virtual reality creates:
   A. A simulated environment
   B. A printed page
   C. A camera lens
   D. A text font

4. Which is a benefit of multimedia in education?
   A. It combines visuals and sound
   B. It replaces all teachers
   C. It uses only text
   D. It removes all interaction

5. HTML is used to:
   A. Compress video
   B. Model 3D objects
   C. Structure web content
   D. Store audio samples

6. Which colour model is subtractive?
   A. CMYK
   B. HSV
   C. RGB
   D. HSL

7. Hypermedia is best described as:
   A. A single video file
   B. Multimedia with hyperlinks
   C. Text-only media
   D. A printed manual

8. MPEG is:
   A. A storage device
   B. A video compression family
   C. A font
   D. An image editor

9. A storyboard is:
   A. A sound file
   B. A video codec
   C. A visual sequence plan
   D. A colour model

10. Interactivity in multimedia means:
    A. Only text is shown
    B. The user can control content and flow
    C. The system plays automatically
    D. There is no navigation

11. A licence gives:
    A. Ownership of all works
    B. Free hardware
    C. Unlimited bandwidth
    D. Permission to use a work

12. CMYK is used mainly for:
    A. Printed output
    B. Screen display
    C. Sound mixing
    D. Animation

13. MIDI stores:
    A. Musical notes and instructions
    B. Digitized audio samples
    C. Video frames
    D. Images

14. Which is NOT a multimedia element?
A. Compiler
B. Animation
C. Audio
D. Text

15. A presentation where the user controls the sequence is called:
A. Non-linear
B. Printed
C. Static
D. Linear

16. A browser is used to:
A. Print documents only
B. Scan images
C. Edit audio
D. Display web pages

17. Chroma key is used to:
A. Scan images
B. Add text
C. Replace a colour background
D. Compress audio

18. A hyperlink links to:
A. Only audio
B. Another resource
C. Only images
D. Only text

19. A serif font has:
A. Only uppercase letters
B. No spacing
C. Small strokes at character ends
D. No strokes

20. Convergence in multimedia refers to:
A. Recording audio only
B. Printing images
C. Deleting media files
D. Merging of media and computing

21. Which is an input device?
A. Monitor
B. Printer
C. Plotter
D. Scanner

22. A bitmap image is made of:
A. Pixels
B. Text characters
C. Mathematical objects
D. Sound samples

23. Which is an output device?
A. Plotter
B. Scanner
C. Keyboard
D. Mouse

24. Which image type scales without losing quality?
A. Scan
B. Photograph
C. Bitmap
D. Vector

25. Kerning adjusts:
A. Space between individual character pairs
B. Line length
C. Font colour
D. Page size

26. Streaming allows media to:
A. Only play offline
B. Play while downloading
C. Remove compression
D. Require full download first

27. Target audience refers to:
A. Colour model
B. Intended users
C. File format
D. Software licence

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28. Plug and play means:
A. Manual driver installation only
B. Automatic device recognition
C. No power needed
D. Wireless only

29. Path animation moves an object:
A. Randomly
B. Along a predefined route
C. Only vertically
D. Without motion

30. Anti-aliasing is used to:
A. Smooth jagged edges
B. Compress video
C. Add sound
D. Increase file size only

PART D (40 MARKS)

QUESTION 1

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(5 marks)

QUESTION 2

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 3

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(2 marks)

b) Describe TWO factors that affect image quality.

(4 marks)

c) Explain how colour model and resolution affect screen and print output.

(4 marks)

QUESTION 4

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

END OF QUESTION PAPER